



Review Article

Mechanisms of radiation-induced endothelium damage: Emerging models and technologies



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ABSTRACT

Radiation-induced endothelial/vascular injury is a major complicating factor in radiotherapy and a leading cause of morbidity and mortality in nuclear or radiological catastrophes. Exposure of tissue to ionizing radiation (IR) leads to the release of oxygen radicals and proteases that result in loss of endothelial barrier function and leukocyte dysfunction leading to tissue injury and organ damage. Microvascular endothelial cells are particularly sensitive to IR and radiation-induced alterations in endothelial cell function are thought to be a critical factor in organ damage through endothelial cell activation, enhanced leukocyte-endothelial cell interactions, increased barrier permeability and initiation of apoptotic pathways. These radiation-induced inflammatory responses are important in early and late radiation pathologies in various organs. A better understanding of mechanisms of radiation-induced endothelium dysfunction is therefore vital, as radiobiological response of endothelium is of major importance for medical management and therapeutic development for radiation injuries. In this review, we summarize the current knowledge of cellular and molecular mechanisms of radiation-induced endothelium damage and their impact on early and late radiation injury. Furthermore, we review established and emerging *in vivo* and *in vitro* models that have been developed to study the mechanisms of radiation-induced endothelium damage and to design, develop and rapidly screen therapeutics for treatment of radiation-induced vascular damage. Currently there are no specific therapeutics available to protect against radiation-induced loss of endothelial barrier function, leukocyte dysfunction and resulting organ damage. Developing therapeutics to prevent endothelium dysfunction and normal tissue damage during radiotherapy can serve as the urgently needed medical countermeasures.

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Abbreviations: ARS, Acute radiation syndrome; AJ, Adherens junctions; ATM, Ataxia telangiectasia-mutated; bMFA, biomimetic microfluidic assay; CNS, Central nervous system; CVD, Cardiovascular disease; DAMP, Damage-associated molecular patterns; G-CSF, Granulocyte-colony stimulating factor; GM-CSF, Granulocyte/Macrophage Colony Stimulating Factor; GI, Gastrointestinal; GGA, Geranylgeranylacetone; HUVEC, Human umbilical vein endothelial cells; ICAM-1, Intercellular adhesion molecule 1; IL-1, Interleukin-1; IR, Ionizing radiation; EC, Endothelial cell; JAK1, Janus kinase 1; JAM-C, Junctional adhesion molecule-C; LET, Linear energy transfer; MHC, Major histocompatibility complex; M-CSF, macrophage colony-stimulating factor; MnSOD, Manganese-dependent superoxide dismutase; MPS, Microphysiological systems; NETS, Neutrophil extracellular traps; PAI-1, Plasminogen activator inhibitor type 1; PAF, Platelet activating factor; PDGF, Platelet-derived growth factor; PKC δ , Protein Kinase C-delta; ROS, Reactive oxygen species; SRS, Stereotactic radiosurgery; TEER, Transendothelial electrical resistance; TGF β , Transforming Growth Factor-Beta; TM, Thrombomodulin; TJ, Tight junctions; TBI, Total body irradiation; TNF- α , Tumor necrosis factor- α ; TYK2, Tyrosine Kinase 2; WBC, White blood cells; WMD, Weapons of mass destruction.

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Healthy tissue may be exposed to ionizing radiation (IR) during radiotherapy [1], nuclear accidents, or by weapons of mass destruction (WMD) [2]. IR produces oxidative stress resulting in acute and chronic cellular damage. The vascular endothelium, which plays an important role in organ homeostasis, is a key target of radiation damage. Microvascular endothelial cells (EC), in particular, are sensitive to radiation and radiation-induced alterations in EC structure and function. Damage to the endothelium is an important regulator of radiation damage in both targeted radiotherapy and whole-body irradiation resulting from exposure to WMD [3]. However, the signaling pattern in these two cases may be different due to, for example, the differential upregulation of cytokines in local vs. whole body irradiation. IR-induced activation of EC leads to enhanced leukocyte-EC interactions, increased permeability, and initiation of apoptotic pathways [4,5]. Radiation-induced endothelial damage and its associated vascular changes often lead to chronic lesions when organs at risk (e.g. lung, kidney, heart and

brain) are exposed to sufficiently high doses [3], and patients with IR-induced tissue damage may die of organ failure.

Moreover, exposure to IR can accelerate atherosclerosis adversely affecting normal tissues resulting in coronary artery disease, peripheral vascular disease, radiation pneumonitis and fibrosis, and cerebrovascular disease. This effect can be increased with higher doses per fraction [6,7]. This is clinically relevant as contemporary treatments move increasingly towards hypofractionation [8,9] in treatments such as stereotactic brain and body radiosurgery (single fraction) and radiotherapy (typically 3–5 fractions) for primary and metastatic brain tumors, early-stage non-small cell carcinoma of the lung and breast conservation irradiation. The radiation target volumes for these treatments must include the planning target volume plus a margin which includes some surrounding normal tissue as defined in the ICRU reports 50 [10], 62 [11] and 83 [12]. This normal tissue can be dose limiting potentially compromising the target volume or resulting in increased short- and long-term secondary effects from EC damage.

Monitoring these vascular damages have become greatly helpful for diagnosis purposes. Recently, many novel methodologies have been developed to monitor the vascular damages. Angiography, magnetic resonance angiography (MRA), computed tomography angiography (CTA) and ultrasound are used clinically for imaging diseases that involve endothelial dysfunction resulting from an inflammatory response [13]. Compared to other techniques, MRA has the advantage of providing high resolution images of the vessel wall. In addition, the recent development of targeted micro-sized particles of iron oxide (MPIO) targeting intercellular adhesion molecules like ICAM-1, VCAM-1, P-selectin and E-selectin have significantly improved the sensitivity and specificity of molecular MRA for imaging endothelial activation compared to other techniques such as plasma biomarkers [14].

Protecting the endothelium protects tissue from radiation damage [15] and therapeutics that specifically prevent EC dysfunction ultimately protect tissue from radiation injury [4] thereby reducing side effects of radiation therapy. Given the central role of the endothelium in normal vascular function, a better understanding of mechanisms of radiation-induced dysfunction is vital in developing therapeutics for radiation-induced endothelium damage [3].

Endothelial dysfunction in irradiated tissue

Normal endothelial cell function

The vascular endothelium is composed of a single layer of cells lining all blood vessels, which acts as a semipermeable barrier regulating the delivery of nutrients, oxygen and cellular components to local tissues and the removal of carbon dioxide and waste products (Fig. 1). The EC barrier regulates the transport of molecules between vascular and tissue compartments through cell–cell junctions and vesicular transport [16]. The endothelium produces several vasoactive substances such as nitric oxide that are vital in regulation of vascular control and growth [17]. With its secretory, synthetic, metabolic, immunologic and surface expression functions, the endothelium plays a critical role in many physiological processes, including control of vascular tone, trafficking of the blood cells between vasculature and underlying tissue, regulation of immune responses, permeability, and angiogenesis [18]. EC show significant heterogeneity in structure, function, cell morphology, gene expression and antigen composition across organs and species [18,19]. This heterogeneity is in part dictated by intrinsic signals as well as the organ-specific microenvironment leading to distinct differences in barrier properties and interactions with circulating immune cells that result in differential sensitivities and responses to radiation.

Endothelial cell dysfunction in early and late radiation damage

Radiation-induced endothelium injury can manifest acutely/early or late after IR exposure. Acute/early radiation syndrome (ARS) encompasses three types of ‘radiation sicknesses’ in bone marrow, gastrointestinal (GI), and cardiovascular/central nervous system (CNS) [20]. GI-ARS is attributed to the radiation sensitivity of the microvascular EC [21] and endothelial dysfunction and increased inflammatory mediators inhibit the recovery of the villus epithelium leading to the breakdown of the epithelial barrier [3]. Moreover, fluid and nutrient absorption across EC is reduced, and mucosal ulcerations develop, resulting in diarrhea, nausea, abdominal pain, and mucous discharge. If ulceration continues, it can lead to bacterial translocation and systemic inflammation [22].

Late vascular effects such as capillary collapse, thickening of basement membrane, scarring and fibrosis may occur weeks to months post-irradiation. Depending on the radiation dose, late pathological syndromes may be delayed in expression, chronic in nature, and associated with evolving pathologies within multiple organ systems, including the bone marrow, GI tract, lung, heart, kidney and the CNS. In the brain, up to 3 weeks after single dose of 15–30 Gy, leukocyte–EC interactions were upregulated [23]. Although hyperpermeability to albumin was observed in the mesentery up to 6 hours after radiation exposure [24] and inhibition of intercellular adhesion molecule (ICAM-1) blocked the hyperpermeability response to radiotherapy, the increase in leukocyte–EC interaction and hyperpermeability can persist for days to weeks after radiotherapy [25]. The hyperpermeability of injured vessels facilitates extravasation of fibrinogen into the extravascular space and fibrinogen undergoes fibrinopeptide cleavage by thrombin, which then results in crosslinking of fibrin by enzymes such as tissue transglutaminase [26,27]. Crosslinking of tissue transglutaminase activates latent TGF β , which further induces tissue fibrosis by promoting deposition of collagen [28].

Ionizing radiation upregulates endothelial cell mediated inflammatory response

IR-induced cellular damage triggers apoptosis and cell death leading to the release of damage-associated molecular patterns (DAMP) and activation of the systemic inflammatory response producing significant changes in the microvascular network structure and function [29–32]. EC acquire a pro-inflammatory phenotype leading to increased cytokine release (Table 1), ROS production, and enhanced EC adhesion molecule expression resulting in increased recruitment of immune cells of myeloid and lymphoid origin [6,24,33–35] (Fig. 1). Increased recruitment of neutrophils, monocytes, and macrophages, as well as Th1 and Th17 lymphocytes, contribute to IR-induced inflammation [6,33,36,37]. This inappropriate influx of immune cells across the vascular endothelium, initiation of vascular EC damage and loss of barrier function have been implicated in the pathogenesis of radiation injury and organ dysfunction [38].

A key step in IR-induced organ damage is excessive adhesion to and migration of activated leukocytes across vasculature (Supplementary Data Video 1 and Video 2) and a reduction in neutrophil infiltration is associated with better outcomes following skin irradiation [4]. Neutrophil recruitment is a multi-step process, which requires crosstalk between neutrophils and EC and includes five discrete steps: capture/attachment, rolling, firm arrest, spreading and extravasation/migration and each step requires crosstalk between leukocytes and EC [39] (Figure 2). Ultimately, arrested neutrophils extravasate to inflamed tissues across EC regulated by concurrent chemoattractant-dependent signals, adhesive events and hemodynamic shear forces [39].

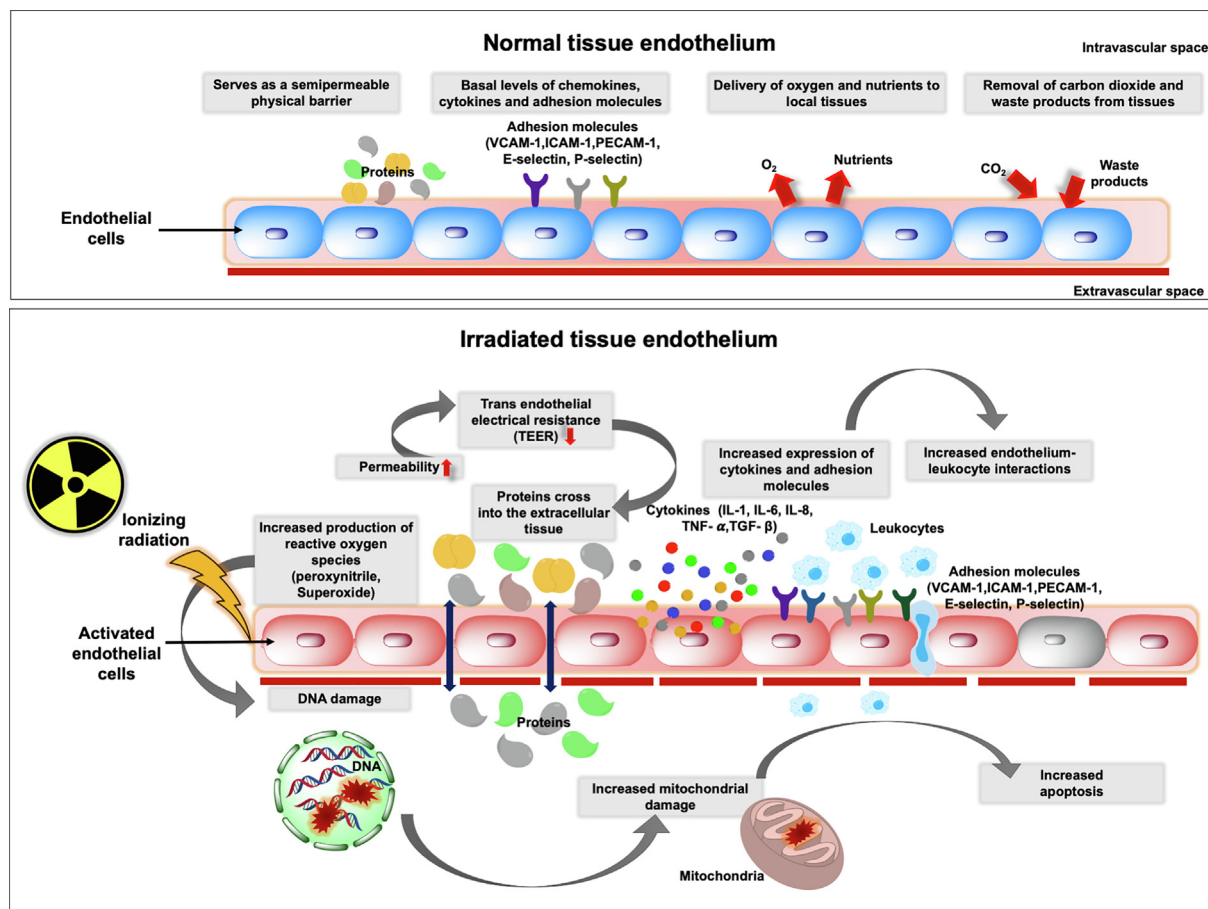


Fig. 1. Overview of vascular endothelium and mechanisms by which IR impacts endothelial cell regulation. Top panel: Normal tissue - Endothelial cells act as a semipermeable barrier that regulates the delivery of nutrients and oxygen to tissue and the removal of carbon dioxide and waste products. Normal endothelial cells have basal levels of some adhesion molecules. Bottom panel: Irradiated tissue - Ionizing radiation increases the production of ROS leading to DNA and mitochondrial damage and increased apoptosis. IR also alters endothelial permeability by acting on tight and adherens junctions allowing excess extravasation of proteins to cross into the extracellular tissue. Radiation exposure also increases the release of proinflammatory cytokines and chemokines and upregulation of adhesion molecules resulting in increased leukocyte-endothelial cell interaction and trafficking to vital organs.

A number of molecules involved in the leukocyte adhesion cascade are also involved in radiation-induced tissue damage. For example, we demonstrated by intravital microscopy that adhesion molecules (ICAM-1, E-selectin) were upregulated in irradiated tissue *in vivo* and the resulting increased leukocyte adhesion was modulated with administration of an anti-ICAM-1 antibody [34,35,40]. We and others have shown that irradiation of EC *in vitro* significantly increased the expression of the adhesion molecules VCAM-1 and ICAM-1 [41,42].

EC also synthesize the neutrophil agonist, platelet activating factor (PAF) [43], which is co-expressed with P-selectin within minutes, while it takes a few hours to co-express E-selectin with IL-8. These co-expression pairs, in turn, upregulate integrin expression on leukocytes. Administration of a PAF antagonist, BN52021, blocked the upregulation of leukocyte adhesion [43].

Neutrophils activated in response to IR-induced systemic inflammation can release neutrophil extracellular traps (NETs) [53]. NETs are composed of nuclear chromatin filaments studded with histones and granular proteins, such as neutrophil elastase and myeloperoxidase. While important for bactericidal activities, NETs can also damage EC and further upregulate the inflammatory response [54,55]. Ability of peripheral blood neutrophils from cervical cancer patients to form NETs increased from 53.6% before radiation therapy to 66.7% after radiation therapy [56] but the mechanisms by which IR activates NETs and the interconnection

between NETs and radiation-induced EC damage have not been fully delineated.

IR-induced endothelium activation increases barrier dysfunction

Damage to and/or the denudation of EC after radiation exposure produces changes in permeability of EC. The vasculature becomes leaky within hours post-IR but the degree to which EC of various vessel types become permeable varies *in vitro* [57]. IR also increases the permeability of the blood-brain-barrier allowing for ionic movement, excess extravasation of inflammatory cells, proteins and biologic response molecules (e.g. growth factors, cytokines) into the brain parenchyma causing brain damage [58,59]. IR alters endothelial permeability by acting on tight and adherens junctions [60]. Tight junctions adhesion is mediated by the claudins family of proteins, which are connected to the cytoskeleton by tight junctions proteins while adherens junctions are formed by classical cadherins that are linked to the cytoskeleton by proteins belonging to the catenin family [61,62]. Among these, vascular endothelial-cadherin (VE-cadherin, a substrate of ADAM10), an important regulator of vascular integrity, which when activated, increases EC permeability [63]. *In vitro*, IR activates ADAM10 and cleavage of VE-cadherin, leading to increased human endothelial permeability to macromolecules of various sizes in a radiation dose dependent manner [60]. Consistent with

Table 1
Cytokines in radiation-induced inflammatory response.

Classification	Cytokines	Response after Radiation Exposure	References
Interleukin and TNF Family	IL-1- β , IL-1- α , IL-8, IL-6, TNF- α , IL-4, IL-13	• Plays a role in generating reactive nitrogen and oxygen species such as nitric oxide or hydroxyl radicals • Induces proto-oncogene expression post-radiation • Promotes T-cell differentiation towards T-helper types 1, 2 and 17 • Initiates neutrophil adhesion, migration and extravasation into tissues • Promotes pro-inflammatory transcription factors such as NF-κB and AP-1 • Enhances vascular adhesion molecule (CAMs, cadherins etc.) expression	[44–49]
Type I and II Interferon family	IFN- α , IFN- β , IFN- γ	• Expressed on WBCs, signal TYK2 and JAK1 and activates STATs. Activation of STATs leads to IL-1β phosphorylation and IL-6 production, leading to inflammation. Activation of STATs can also inhibit cell division, activate and drive WBC formation in inflammation. • Leads to expression of MHC class II on endothelial and immune cells	[47,50]
Colony stimulating factor family	G-CSF, GM-CSF, M-CSF	• Induces WBC production, leading to increased trafficking of leukocytes in response to inflammation	[47]
Growth factor family	TGF β , PDGF	• Increases adhesion molecule expression on EC • Increased expression leading to increased collagen production and tissue remodeling • Strong stimulators of fibrosis	[51,52]

other studies [63], we demonstrated that exposure of human EC under shear flow to IR in a novel microphysiological system significantly increases permeability and decreases transendothelial electrical resistance (TEER) across the endothelial barrier [41].

Radiation exposure leads to mitochondrial dysfunction

In many mammalian cells, mitochondria produce cellular energy but mitochondrial content in EC is limited and EC ATP production is primarily via aerobic glycolysis [64]. When HUVECs were exposed to 5–20 Gy of IR, the mitochondrial membrane potential decreased and mitochondrial ROS production increased [65]. Furthermore, murine cardiac microvascular EC acquired protein expression profiles related to mitochondrial dysfunction when they were irradiated with 8 and 16 Gy X-rays [66]. It is hypothesized that EC mitochondria functions more as important signaling organelles [67] and the three main mitochondrial functions, Ca²⁺ regulation, control of cell death, and oxidative stress signaling are disturbed following IR exposure [68]. Mitochondrial Ca²⁺ signaling is altered by IR [69], and at IR doses high enough to overcome cellular antioxidant responses, oxidative stress can lead to mitochondrial dysfunction [70].

IR increases apoptotic cell death

Apoptotic cell death due to radiation exposure can be primarily mediated by either p53 or the sphingomyelin/ceramide pathways [6]. It's shown that p53 is important in response to DNA damage, which activates mitochondrial-mediated cell apoptosis [71]. Endothelial cell apoptosis at doses > 5 Gy can be induced by persistent DNA damage, resulting in leading to p53 accumulation and resulting in activation of the caspase pathway, [72] but mechanisms of endothelial cytotoxicity at lower radiation doses are less known [73]. It has been proposed that EC apoptosis after high-dose single-fraction IR is primarily modulated by the sphingomyelin ceramide pathway [74]. Sphingomyelin is a phospholipid present in the cell membrane which is hydrolyzed by TNF activation after IR exposure. Extracellular acidic sphingomyelinase then leads to radiation-induced, ceramide-mediated EC apoptosis [75]. A classical study has shown the requirement of the sphingomyelinase encoding gene, ASMase, for mediating ceramide generation in induction of microvascular endothelial apoptosis [21] and high dose IR triggering of ASMase/ceramide signaling in EC has been shown to play a major role in membrane signaling after radiation exposure [76]. However, the precise relationship between IR dose and EC apoptosis has not been clearly established [77].

IR also induces differential effects on microRNA (miRNA) levels in EC. *In vitro* studies demonstrated that miRNAs have a role in EC

clonogenic survival and cell growth, as well as impacting EC radiosensitivity indicating an important role of miRNAs in the EC response to radiation [78].

Emerging models used for studying the effects of IR on endothelium

In vivo animal models

Although no single animal model can completely represent the human condition, *in vivo* models including mouse, rat, rabbits, and pigs have been widely used to study radiation-induced endothelium damage [68]. The choice of an appropriate animal model for a specific study is generally dictated by factors such as anatomy, physiology, genetics, and immune response of the animal model being considered. Mice show similarities to human physiological and genome (>95%) [79] and have been used widely to study radiation induced upregulation of cell adhesion molecules, leukocyte-endothelial interaction and EC damage [34,80]. However, the mouse may not be the optimum animal model for studying radiation induced damage for multiple reasons. A major disadvantage is their small body thickness, which does not account for the heterogeneity of radiation dose distribution characteristic to human exposure [79]. Additionally, despite the phylogenetic relatedness, translating studies from mouse immune system to human disease is complicated by many factors including size, metabolic rate, WBCs composition, and differences in pathophysiology and drug LD₅₀ values that are significantly different from humans [3,81]. Given the complexities inherent in the use of animal models, several *in vitro* systems have been developed that use human cell-based assays to provide a better understanding of radiation-induced cell damage in humans.

In vitro cell culture systems

EC grown as a homogeneous 2D monolayer are traditionally used to complement animal models. A number of these *in vitro* EC models have been used to study effects of IR on activation of the NF κ B pathway [82], alterations in mitochondrial membrane potential [65] and the effect on vascular tone [83]. The effect of IR on EC permeability and barrier function can also be monitored in 2D models through measurement of TEER, EC adherens junctional integrity, and EC monolayer permeability [84]. Although using primary EC can provide important insight into various physiological and pathological processes, cell culture presents significant challenges. For example, almost all primary EC have an average life span of 5–10 serial passages *in vitro*, after which these cells often stop proliferating when they form multinucleated cells

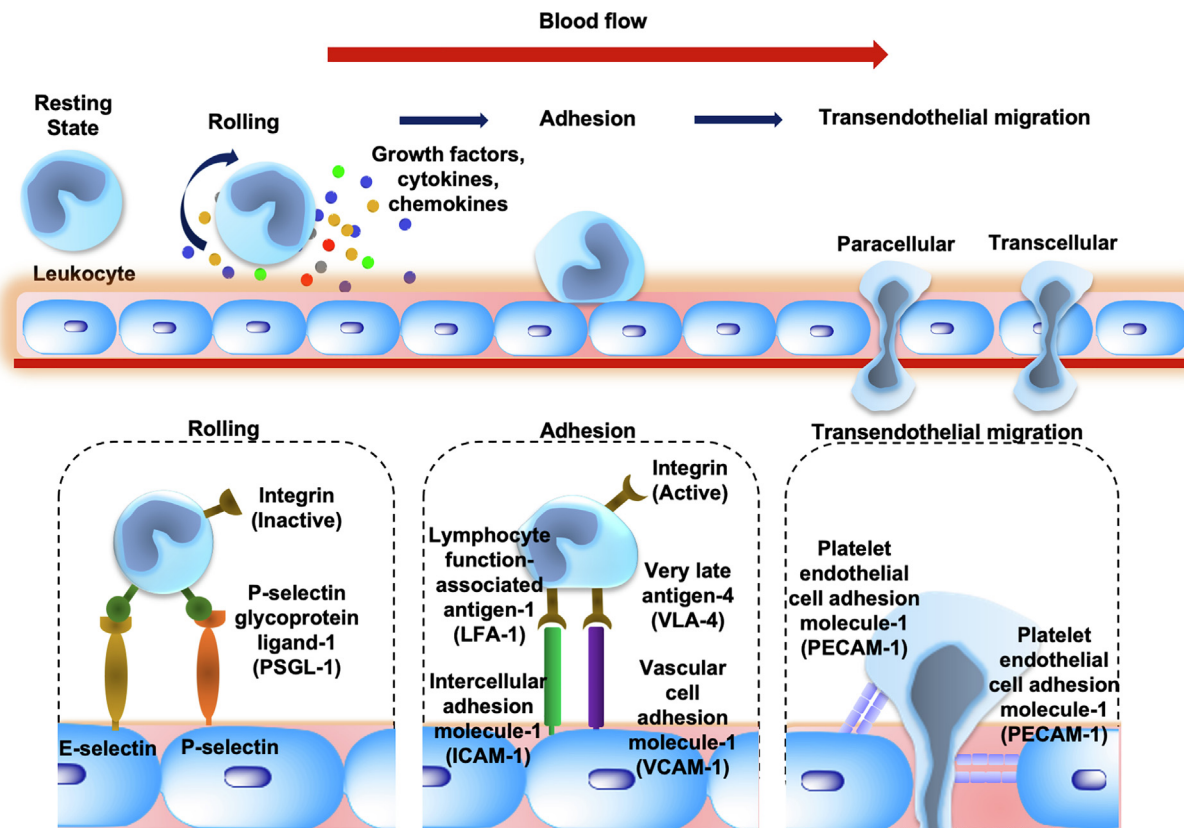


Fig. 2. Multi-step process of neutrophil recruitment that includes rolling, adhesion and transmigration. On endothelial cells, selectins (e.g., P- & E-selectin) are responsible for neutrophil capture and rolling, while adhesion molecules ICAM-1, VCAM-1 and PECAM-1 are critical regulators of neutrophil firm attachment and migration.

and eventually die [85]. Another drawback is the phenotypic differences between large vessel-derived EC (e.g. HUVEC) vs. EC of microvascular origin (e.g., human dermal microvascular cells) but also between EC derived from different organs. These mostly static cell culture systems fail to reproduce many of the complexities of the *in vivo* conditions including three-dimensional EC morphology, cell–cell and cell–matrix interactions, and shear flow conditions [86]. Nevertheless, these low cost, easy to use, and high-throughput systems are considered standard for analysis of cellular responses [86]. Flow-based 2D cultures offer advantages as they have more precise spatial/temporal dynamics, sensor integration and continuous monitoring [35,86].

Microphysiological systems

The inability of 2D monolayer cultures to recreate the appropriate microenvironment for cell–matrix, cell–cell, cell–tissue and cell–organism interactions has led to the development of an emerging new class of *in vitro* models that better mimic the *in vivo* conditions. These new models not only provide novel approaches for studying radiation damage to EC but also offer a new tool for the rapid development and screening of therapeutics. Microphysiological systems, often synonymous with “organ-on-chip”, commonly consist of an interconnected set of 2D and 3D cell cultures in microfluidic devices. These emerging organ-on-a-chip systems can increasingly recapitulate the complex microenvironment of different human tissues by recreating organ-specific geometries and co-culturing of multiple cell types, mimicking the physiological and pathological conditions and responses to therapeutics. Microphysiological systems can serve as *in vitro* models of brain, GI tract, lung, liver, vasculature, and skin microvasculature [87] and are rapidly becoming the new standard tool for better understanding vascular biology, pharmacology and toxicology [87]. Sev-

eral microphysiological systems have been used to study the effects of radiation on different organs and screen various therapeutics [41,88]. Employing both human and mouse cells, together with omic and *in silico* modeling approaches, will help better predict how therapeutics developed in animal models may work in clinical settings as well as screen potential therapeutics on patient cells for personalized medicine [89].

Organoids, spheroids, and scaffolds

Recent advances in tissue engineering have facilitated the development of novel 3D models for radiobiological research including organoids, spheroids and scaffolds. Organoids consist of a collection of organ-specific cell types that spatially organize to restrict lineage commitment similar to *in vivo* [90]. Spheroids are aggregated, mutually adherent population of cells with spherical shapes [91]. Although organoids have *in vivo* like cellular architecture and spheroids have advantages of high reproducibility [92], they are still not widely used in studying radiation-induced EC damage in part due to the major challenges associated with developing organoids and spheroids. A key limitation in using organoid and/or spheroid based approaches to generate functional tissue is that upon reaching a certain size, organoids switch from a proliferative, stem-like state to a non-proliferative, terminally differentiated one and develop a necrotic core [93]. With the exception of a few avascular tissues, any efforts at producing tissues would be limited to a length scale of approximately 150 μm , the natural diffusion limit of oxygen in tissue. However, scalable 3D microfabrication technologies like layer-by-layer deposition of materials and selective removal of materials to form tubular voids, as well as a range of hybrid approaches utilizing sacrificial materials [93], allow for generation of free-form vascular structures in organoids [94,95].

Scaffold/matrix-based 3D cultures (Fig. 3A) are formed by seeding cells on a 3D synthetic-based matrix or by dispersing cells in a liquid matrix followed by polymerization to mimic the tissue microenvironment [92]. Scaffolds have high reproducibility and allow for coculture of other cell types but frequently have a simplified architecture [92]. While 3D cultures are considered more informative for studying radiation damage, they are not as easy to propagate and standardize as monolayer cultures. These 3D *in vitro* models must incorporate both the spatial organization and differentiated function of the tissue *in vivo* to allow reproduction of cell–cell interactions and create the appropriate microenvironment for cellular proliferation and gene expression [96]. For example, a 3D capillary model using human umbilical vein EC (HUVEC) grown in a collagen gels showed more complex and persistent 53BP1 DNA damage foci when exposed to high LET (linear energy transfer) as compared to low LET [97]. However, due in part to the complexity and lack of reproducibility of some 3D models, more realistic *in vitro* assays using human cells are needed to better mimic the *in vivo* microenvironment.

Microfluidic systems

Microfluidic systems (Fig. 3B) permit the study of complex vascular and microvascular processes, such as neutrophil–EC interactions and the inflammatory response, using *in vitro* 3D models that more realistically reproduce the tissue microenvironment. For example, Gut-on-a-chip models have been used to study the effects of gamma (γ)-radiation on villus intestinal epithelium [88]. Exposure to γ -radiation increased the generation of ROS, cell cytotoxicity, apoptosis, and led to compromised intestinal barrier integrity. A microvasculature-on-a-chip was used to investigate the effects of IR on HUVEC forming 3D perfusable networks mimicking the human microvasculature [98]. Importantly, a systematic comparison between HUVEC cultured in this system with a traditional 2D monolayer model showed significant differences upon IR exposure, particularly at high doses that are typically used in stereotactic body radiation therapy (SBRT) and stereotactic radiosurgery (SRS). At high IR doses up to 25 Gy, VE-cadherin cell–cell adherens junctions did not change significantly in the 3D fluidic model but were significantly damaged in the 2D monolayer model. Similarly, increased apoptosis was observed in the 2D monolayer model as compared to the 3D fluidic model. An advantage of the 3D models is that ECs form 3D tube-like structures in these model systems mimicking a more realistic *in vivo* environment and behave quite differently than those cultured in a 2D environment [98]. The fact that many of the *in vitro* studies of effects of IR has traditionally resulted from studies conducted with clonogenic assays in a 2D environment should give impetus to validate these findings in microfluidic systems mimicking the 3D *in vivo* microenvironment.

Our group has developed a novel biomimetic microfluidic assay (bMFA) that facilitates real-time assessment of neutrophil rolling, firm arrest, spreading and migration to the extravascular tissue in a realistic microvasculature geometry under physiologic shear conditions in a single assay (Fig. 3 C–F). This is one of the few *in vitro* systems that realistically model *in vivo* geometrical features, including vascular morphology and flow conditions such as converging or diverging flow at bifurcations of the vasculature [41,99–101]. This integrated microfluidic assay provides a novel platform for investigating the EC inflammatory response, neutrophil–EC interaction, and EC damage in response to various stimuli including IR (Supplementary Data, Video 3). We have used bMFA to demonstrate that IR exposure upregulated human neutrophil–EC interaction, increased ICAM-1 and VCAM-1 (but not E-selectin) expression, and increased permeability under shear flow conditions [41].

Therapeutics

Radioprotective agents are classified into three broad categories: radioprotectors, radiomitigators and radiotherapeutics. Radioprotectors are administered prior to radiation exposure to protect downstream injury and include molecules with thiol functionalities and antioxidant properties. Radiomitigators are given during or quickly after radiation exposure, primarily to mitigate normal tissue toxicity, stop/alleviate the consequences of tissue exposure to radiation and to hasten repair (Table 2). Radiotherapeutics are given after the clinical appearance of normal tissue toxicity to ameliorate radiation damage and to initiate tissue regeneration or repair [102]. Radiomitigators and radiotherapeutics can also serve as radiation countermeasures during mass casualty events resulting from radiation exposure.

Lack of selective anti-inflammatory therapeutics complicates radioprotection

While the use of Neupogen and Neulasta to treat hematopoietic acute radiation syndrome (H-ARS) was recently approved by the FDA, therapeutic approaches to the treatment of radiation-induced vascular/ endothelial injury are largely supportive and there are no specific pharmacologic therapies available that protect from radiation-mediated tissue damage [3]. Given the fact that radiation damage to EC initiates and is mediated by the immune response, radioprotection is also hampered by a lack of effective anti-inflammatory therapeutics. Most anti-inflammatory therapeutics work primarily through immunosuppression, rather than selective immune modulation. This lack of effective anti-inflammatory agents continues to complicate the treatment of inflammatory pathologies ranging from sepsis and COVID-19 to radiation-induced inflammatory response. Radiotherapies for treatment of IR-induced endothelial damage have been primarily focused in 3 areas, targeting EC pathways activated in response to radiation, reconstitution of bone marrow and reduction in vascular barrier permeability and vascular leak [3].

Radioprotectors

Potential radioprotectors include BAY 11-7085, an inhibitor of NF- κ B activation, has been shown to partially inhibit the effects of γ -radiation on permeability and TEER in HUVECS [103]. Geranylgeranylacetone (GGA), a drug used to treat peptic ulcers and gastritis, has been shown to moderate IR-induced intestinal injury and EC dysfunction [104]. Intestinal microvessels were preserved in GGA-treated mice thereby reducing intestinal injury. GGA plays an important role in the promotion of angiogenic activity in damaged EC via inducing VEGF/eNOS signaling and suppressing inflammatory cytokine expression [104].

Radiomitigators/radiotherapeutics

Pentoxifylline is a radiomitigator/radiotherapeutic used to treat vascular disease by inhibiting platelet coagulation, enhancing red blood cell deformability and promoting vessel blood flow. Pentoxifylline also inhibits neutrophil adhesion to EC by decreasing platelet coagulation via platelet activating factor (PAF). Administering pentoxifylline and another alpha tocopherol/vitamin E to patients undergoing radiotherapy lowered their pulmonary toxicity compared to controls after six weeks [105].

Endothelial specific radiomitigators/radiotherapeutics

Several potential therapeutics that target IR-induced damage to EC are currently being evaluated in *in vitro* cell cultures and animal

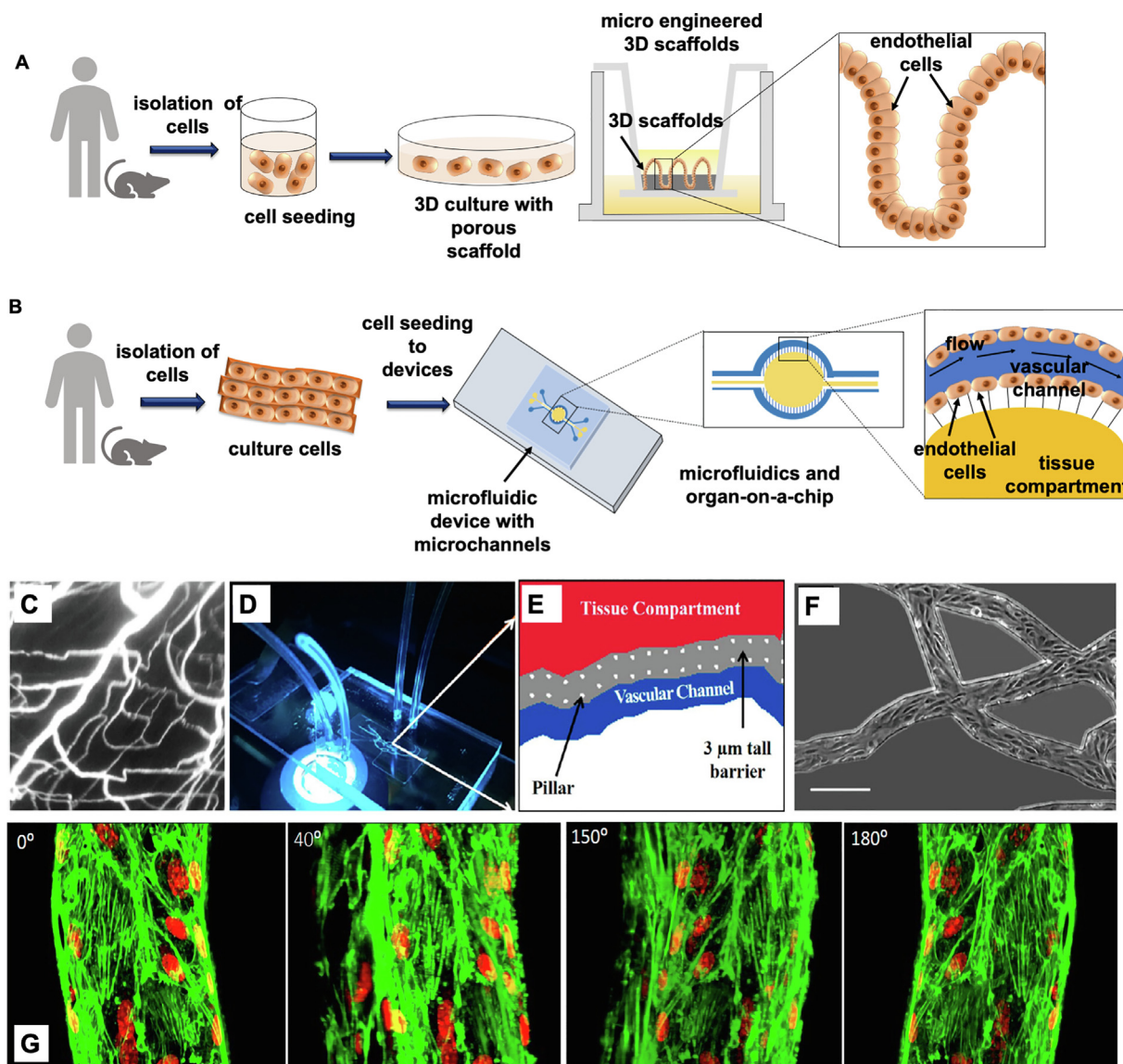


Fig. 3. Use of microphysiological systems in radiobiological research: (A) micro engineered 3D scaffolds; (B) a novel *in vitro* biomimetic microfluidic assay (bmFA) developed to study radiation-induced endothelium damage; (C) map of microvascular networks in animals obtained using intravital microscopy; (D) vascular network reproduced on polydimethylsiloxane device; (E) the bmFA includes vascular channels that are connected to the tissue compartment through a 3 µm barrier; (F) EC are aligned in the direction of flow in the bmFA (scale bar 250 µm); (G) confocal microscopy demonstrates that EC form a complete 3D lumen in the vascular channel. F-actin is labeled in green, and nuclei are labeled in red. [C–G: reproduced with permission from reference [41]].

models. A novel tetrapeptide N-acetyl-Ser-Asp-Lys-Pro (Ac-SDKP) has been shown to localize to EC and inhibit IR-induced endothelial ROS generation [106]. *In vivo* studies demonstrated that Ac-SDKP administration was able to restore IR-induced endothelial barrier damage by preventing EC cell loss and increasing expression of tight junction proteins. Ac-SDKP also showed strong antifibrotic effects that inhibited IR-induced coronary vascular fibrosis thus preserving myocardial blood flow [106]. The statin, Pravastatin, has also been shown to exert persistent anti-inflammatory and anti-thrombotic effects on irradiated EC and to inhibit leukocyte-EC interaction [107]. Pravastatin also regulates thrombomodulin (TM) expression which inhibits the leukocyte adhesion to the irradiated EC. Although the mechanisms of TM transcription activation by this drug is unclear, suppression of IR-induced endothelial interaction with leukocyte may prevent intestinal inflammation post-irradiation [107]. Angiotensin II (Ang II) is produced by the endothelium in response to IR and induces the generation of the profibrotic cytokine TGF- β . *In vitro* treatment of EC with the ACE inhibitor, Captopril, following IR was shown to reduce Ang II

expression, inhibit the NF- κ B pathway, and reduce TGF- β 1 expression [108].

Genetic approaches employing knockout mice have identified other EC targets to mitigate IR-induced cell damage. For example, endothelial expression of the plasminogen activator inhibitor type 1 (PAI-1) following IR increases and plays a role in activating endothelial apoptosis. In response to IR, PAI-1^{-/-} mice had increased EC survival, preserved vascular density and mucosal integrity as compared to wild type mice [109]. Mice with key proapoptotic proteins Bak and Bax deletions in both bone marrow EC and bone marrow hematopoietic stem cells (HSCs) exhibited protection of the BM vasculature and increased survival post IR exposure [110].

Emerging endothelium-based radiotherapeutics

To respond to the urgent need for new therapies, new classes of novel therapeutics are being developed to specifically treat endothelial dysfunction resulting from radiation exposure. For

Table 2

Therapeutics developed to treat ionizing radiation induced vascular injury.

Therapeutic	How has it been tested?	Results	Clinical trials? FDA approved?
Neupogen (Radio mitigator)	Nonhuman primates, minipig [119,120]	Enhanced survival, stimulated recovery from neutropenia, induced mobilization of hematopoietic progenitor cells [120]	FDA approved for medical countermeasure to increase survival in patients exposed to myelosuppressive doses of radiation. FDA approved based on animal studies [121]
Neulasta (Radio mitigator) A pegylated form of granulocyte-colony stimulating factor (PEG-G-CSF).	Mouse and nonhuman primates [122,123]	Significant survival efficacy as a single dose in a murine model of the Hematopoietic Acute Radiation Syndrome (H-ARS) [122], Administration at days 1 and 7 was most effective at improving neutrophil recovery compared to daily administration [123]	Phase II clinical trials [124], FDA approved to increase WBC count after chemotherapy [124]
Pentoxifylline (Radiomitigator/ Radiotherapeutic)	Patients [105]	After 6 weeks, patients had lower pulmonary toxicity [105]	Clinical trials with vitamin E to reduce fibrosis [125,126]. FDA approved for the treatment of intermittent claudication on the basis of chronic occlusive arterial disease of the limbs [127]
Endogenous tetrapeptide (Ac SDKP) (Radiomitigators/ Radiotherapeutic)	Rat and <i>in vitro</i> model	Inhibited EC loss, reduced coronary fibrosis, restored TJ-assembly. <i>In vitro</i> , localized to EC and inhibited ROS generation [106]	–
Pravastatin (Radio mitigators/ Radiotherapeutic)	Mouse and <i>in vitro</i> model [107]	Regulate thrombomodulin (TM) expression, exogenous TM inhibits leukocyte adhesion to EC [107]	–
Captopril (Radio mitigator/ Radiotherapeutic)	<i>In vitro</i> model: Human endothelial hybrid cell line EA. HY926 [108]	Reduced Ang II and TGF- β 1 expression and inhibited NF- κ B pathway [108]	FDA approved for the treatment of hypertension [128]
BAY 11-7085 (Radioprotector)	<i>In vitro</i> model: HUVEC [103]	Decrease of TEER, partially blocking permeability increase [103]	–
Geranylgeranylacetone (GGA) (Radioprotector)	Mouse [104]	Ameliorated intestinal injury, preserved intestinal microvessels [104]	–
Plasminogen activator inhibitor type 1 (PAI-1)	WT C57BL/6J (PAI-1 +/+) and PAI-1 –/– mouse [109]	Increased EC survival, vascular density, mucosal integrity [109]	–
Bak and Bax	Chimeric <i>Tie2Bak/Bax^{fl/-}</i> ;WT-EC mouse [110]	Bak and Bax deletion in bone marrow EC, protection of bone marrow vasculature [110]	–
Rosiglitazone (Radioprotector)	<i>In vitro</i> : Human telomerase-immortalized coronary artery EC [112]	Increased oxidative metabolism, redox state, decreased levels of apoptosis [112]	FDA approved for use in management of type 2 diabetes mellitus [129]
PKC δ -TAT peptide inhibitor (Radiomitigator/ radioprotector)	HUVEC in a microfluidic device [41], mice	Reduced radiation-induced EC damage, leukocyte migration [41], reduced mortality in irradiated mice	Safely tolerated in Phase I/II clinical trials for treating acute myocardial infarction [117,118]

example, the peroxisome agonist, Rosiglitazone, which is used clinically to treat diabetes [111], was recently shown to preserve mitochondrial function after radiation exposure and to stimulate mitochondria biogenesis in EC. When coronary EC were treated with Rosiglitazone before IR exposure, the cells exhibited enhanced respiratory function used for ATP production and apoptosis was reduced [112]. The effects of treating EC with Rosiglitazone after radiation exposure have not been reported, so the efficacy of this therapeutic for treating victims of mass/accidental radiation exposure remains unclear.

We identified Protein Kinase C- δ (PKC δ) as a critical regulator of the inflammatory response controlling leukocyte infiltration across endothelium and loss of barrier function. PKC δ is activated in multiple cell types in response to radiation exposure and is involved in radiation-induced apoptosis [113]. Moreover, PKC δ overexpression enhanced radiation-induced apoptosis indicating a critical signaling role [114]. PKC $\delta^{-/-}$ mice were protected from radiation-induced damage to the salivary gland and thymus [115].

We have shown that delivery of a specific PKC δ -TAT peptide inhibitor to the lung had a dramatic anti-inflammatory and lung protective effect in a rodent model of sepsis [116]. Similarly, we have used our novel microfluidic assay (bMFA) to show that PKC δ inhibition dramatically attenuates radiation-induced human EC damage, leukocyte migration (Fig. 4A) as well as vascular EC permeability (Fig. 4B) and preserved integrity of irradiated EC

(Fig. 4C–E) even when administered 24 hrs post-IR [41]. Moreover, neutrophil adhesion to irradiated EC was significantly decreased after PKC δ inhibition in a shear-dependent manner. PKC δ inhibition downregulated radiation-induced P-selectin, ICAM-1 and VCAM-1 overexpression [41].

In preliminary studies, treatment of whole body 7 Gy irradiated C57BL/6J mice with the PKC δ inhibitor significantly delayed and reduced mortality (Fig. 4F), supporting the hypothesis that PKC δ is activated in response to radiation and PKC δ inhibition provides a protective effect. Furthermore, administration of the PKC δ -TAT peptide inhibitor used in our study has already been shown to be safely tolerated in Phase I/II clinical trials for treating acute myocardial infarction [117,118].

In summary, radiation-induced EC injury in large part mediates and regulates wider tissue damage. Ionizing radiation directly activates and damages the endothelium via increased adhesion molecule expression, leukocyte-EC interactions, mitochondrial damage, barrier permeability and apoptosis. Emerging *in vivo* and *in vitro* models of vascular inflammatory response are providing not only a better understanding of mechanisms underlying the progression of radiation-induced EC damage but also a roadmap for developing highly specific radiotherapeutics for preventing and treating side effects of radiotherapy and/or accidental radiation exposure. Novel therapeutics that specifically focus on common key control points or signaling hubs can more effectively regulate the multiple over-

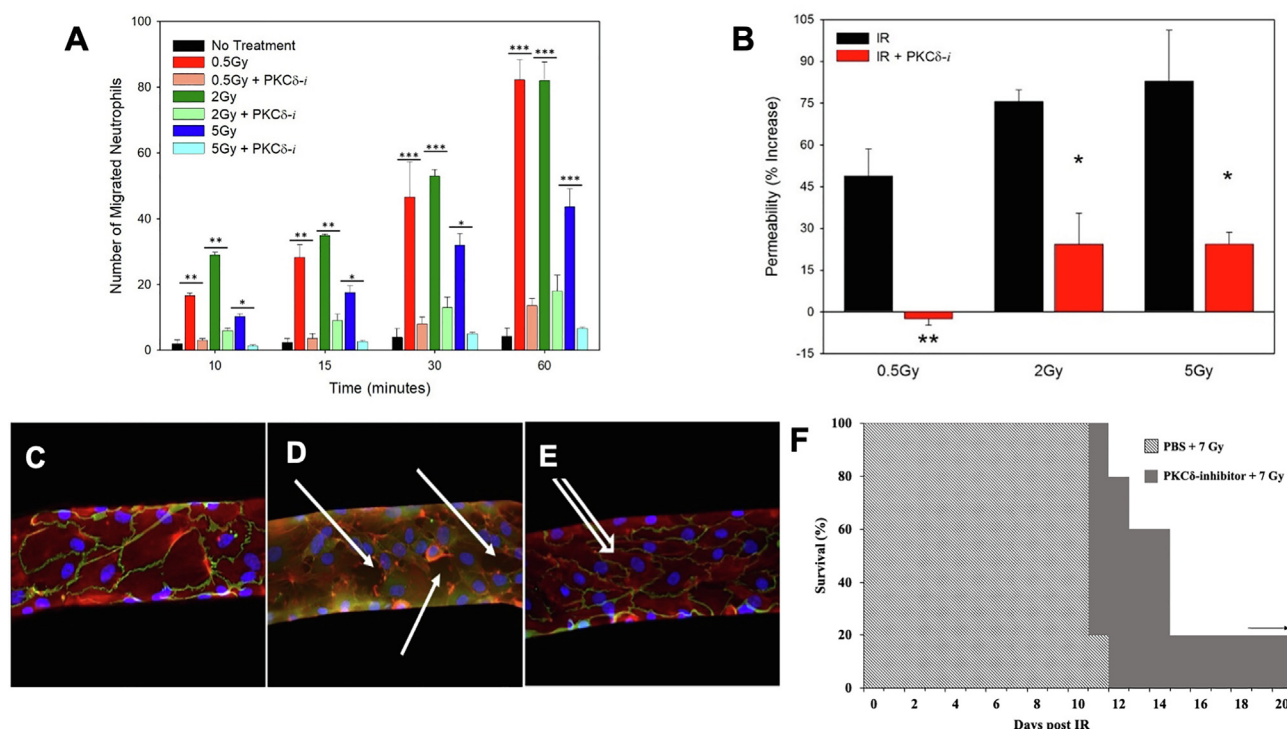


Fig. 4. PKC δ inhibition as a novel medical countermeasure for radiation-induced vascular damage; (A) Neutrophil migration across irradiated human EC increases over time by up to 20-fold at 60 min. PKC δ -TAT inhibitor (PKC δ -i) at 24 hours post-IR significantly reduces neutrophil migration by up to 82% after 60 min; (B) Dextran permeability of EC exposed to irradiation is significantly increased. Treatment of cells with PKC δ -i restores permeability to control levels (0 Gy). Data are normalized with respect to the permeability of EC with no treatment; (Mean $\pm$ SEM, $n = 3$ /group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). (C) EC are aligned in the direction of flow under control conditions; (D) whereas in response to 5 Gy IR, they are not as well aligned and denuded (solid arrows); (E) PKC δ -i 24 hrs post-IR prevents denuding of EC which align in the direction of flow (open arrow); green: VE-cadherin (adherens junction); red: phalloidin (actin filament); blue: Hoechst 33342 (cell nucleus). [Reproduced with permission from reference [41]]; (F) all control mice whole body irradiated with 7 Gy treated with PBS died between days 11 and 12 post-IR, while 80% of mice whole body irradiated with 7 Gy treated with PKC δ -i lived to days 12–16 post-IR, with one mouse living for >60 days ($\rightarrow$) post-IR when it was euthanized as required by our animal protocol ($n = 5$ /group).

lapping and redundant mechanisms that modulate the signaling pathways regulating the vascular endothelial response to radiation-induced damage.

Conflict of interest statements

All authors confirm that they have read and approved the final version of the manuscript being submitted. They also certify that the article is not under consideration for publication elsewhere.

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Appendix A. Supplementary data

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